Transverse field Ising model

$$H = -J\left[\sum_i \sigma_i^z \sigma_{i+1}^z + g\sum_i \sigma_i^x\right]$$

<u>Large-g</u> limit: ground state is eigenstate of $\hat{\sigma}_i^x$

$|0\rangle = \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow$ $\quad H \Rightarrow -Jg\sum \hat{\sigma}_i^x \Rightarrow \Psi_0 = \underbrace{\prod_n \Psi_{1n}}_{\text{product state}}$

Excited $|n\rangle = \rightarrow \rightarrow \underset{n}{\leftarrow} \rightarrow \rightarrow$ $\quad E_1 - E_0 = 2gJ$

(localized)

$$|n\rangle = \prod_{j\neq n} \Psi_{1j} \; \Psi_{2n}$$

$$= \Psi_{1,1}\,\Psi_{1,2} \ldots \Psi_{2,n}\,\Psi_{1,n+1} \ldots$$

Add weak $\sum_i \sigma_i^z \sigma_{i+1}^z$

$$\underset{\begin{pmatrix}0&1\\1&0\end{pmatrix}}{\sigma^x} \begin{pmatrix}a\\b\end{pmatrix} = \pm\begin{pmatrix}a\\b\end{pmatrix} \Rightarrow (1)\ \begin{pmatrix}b\\a\end{pmatrix} = +\begin{pmatrix}a\\b\end{pmatrix} \Rightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} = \Psi_1$$

$$b = a$$

$$(2)\ \begin{pmatrix}b\\a\end{pmatrix} = -\begin{pmatrix}a\\b\end{pmatrix} \Rightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} = \Psi_2$$

Indeed

$$\begin{pmatrix}0&1\\1&0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} \Rightarrow \hat{\sigma}^x\Psi_1 = \Psi_1$$

$$\begin{pmatrix}0&1\\1&0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}-1\\1\end{pmatrix} = -\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} \Rightarrow \hat{\sigma}^x\Psi_2 = -\Psi_2$$

How $\hat{\sigma}^z$ acts on it?

$$\hat{\sigma}^z \frac{1}{\sqrt{2}}\begin{pmatrix}1\\\pm1\end{pmatrix} = \begin{pmatrix}1&0\\0&-1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\\pm1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\\mp1\end{pmatrix}$$

$$\Rightarrow \hat{\sigma}^z\Psi_1 = \Psi_2$$

$$\hat{\sigma}^z\Psi_2 = \Psi_1$$

Then $\hat{\sigma}_j^z\hat{\sigma}_{j+1}^z\,\Psi_0 = \rightarrow \rightarrow \rightarrow \underset{j}{\leftarrow}\ \underset{j+1}{\leftarrow} \rightarrow \rightarrow$ $\quad E_2 = 4gJ$

creates defects on sites $j, j+1$

(flips spins)

$$\hat{\sigma}^z_j \underline{\underline{\hat{\sigma}^z_{j+1}}} \left(\ldots \psi_{1j} \underline{\underline{\psi_{2j+1}}} \psi_{1j+2} \ldots \right) = \underbrace{\left(\hat{\sigma}^z_j \psi_{1j}\right)}_{\psi_{2,j}} \underbrace{\left(\hat{\sigma}^z_{j+1} \psi_{2j+1}\right)}_{\psi_{1,j+1}} \psi_{1j+2} \ldots$$

moves flipped spin to the left!

$$\begin{array}{ccccc} & & j & j+1 & j+2 \\ \rightarrow & \rightarrow & \rightarrow & \leftarrow & \rightarrow \\ \rightarrow & \rightarrow & \leftarrow & \rightarrow & \rightarrow \end{array}$$

Clearly $\hat{\sigma}^z_{j+1} \hat{\sigma}^z_{j+2} \left(\ldots \psi_{1j} \psi_{2j+1} \psi_{1,j+2} \ldots \right) = \psi_{1j} \psi_{1j+1} \underline{\underline{\psi_{2,j+2}}}$
moves it to the right!

Thus $\hat{H} |n\rangle = -J\left(|n-1\rangle + |n+1\rangle\right) + 2gJ\,|n\rangle$

Form a plane wave $|n\rangle = \frac{1}{\sqrt{N}} \sum_{k} e^{ikr_j} |k\rangle$

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_{j} e^{-ikr_j} |j\rangle$$

$$\hat{H} \frac{1}{\sqrt{N}} \sum_n e^{-ikn} |n\rangle = -J\left(\frac{1}{\sqrt{N}} \sum_n e^{-ik(n-1)} e^{-ik} |n-1\rangle \right.$$

$$\left. + \frac{1}{\sqrt{N}} \sum_n e^{-ik(n+1)} e^{ik} |n+1\rangle \right) +$$

$$+ 2gJ \frac{1}{\sqrt{N}} \sum_n e^{-ikn} |n\rangle$$

$$\hat{H} |k\rangle = \left\{ -J\left(e^{-ik} + e^{ik}\right) + 2gJ \right\} |k\rangle =$$

$$= \left\{ 2gJ - 2J\cos k \right\} |k\rangle$$

$$E_k = 2J(g - \cos k)$$

<u>Small-g limit</u> $\quad H \to -J \sum_i \hat{\sigma}_i^z \hat{\sigma}_{i+1}^z$

$$|0_\uparrow\rangle = \uparrow\uparrow\uparrow\uparrow\uparrow \quad \text{or} \quad \downarrow\downarrow\downarrow\downarrow\downarrow\downarrow = |0_\downarrow\rangle$$

$$\uparrow = \varphi_1 = \begin{pmatrix}1\\0\end{pmatrix} \qquad \downarrow = \varphi_2 = \begin{pmatrix}0\\1\end{pmatrix}$$

$$\hat{\sigma}^z \varphi_1 = \begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix} \text{ eigenvalue } +1.$$

$$\hat{\sigma}^z \varphi_2 = \begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\-1\end{pmatrix} = -\begin{pmatrix}0\\1\end{pmatrix} \text{ eigenvalue } -1.$$

$$\hat{\sigma}^x \varphi_1 = \begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix} = \varphi_2, \quad \hat{\sigma}^x \varphi_2 = \begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix} = \varphi_1$$

$$H|0_{\uparrow,\downarrow}\rangle = -J\sum_i \hat{\sigma}_i^z \hat{\sigma}_i^z |0_{\uparrow,\downarrow}\rangle = -J\sum_i (1)\, |0_{\uparrow,\downarrow}\rangle =$$
$$= E_0 \, |0_{\uparrow,\downarrow}\rangle.$$

Excitations : naively, single spin flip

$$\uparrow\uparrow\underset{|n\rangle}{\uparrow\downarrow}\uparrow\uparrow$$

$H|n\rangle = (E_0 + 4J)\,|n\rangle$ $\quad$ two bonds, $(n-1,n)$ and $(n,n+1)$ change from $(+1)$ to (-1)

$\Rightarrow \Delta E = E_1 - E_0 = 2(2J) = 4J.$

However, <u>domain wall</u> has lower energy

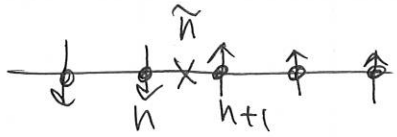
$$\underset{\ldots\, n-2}{\downarrow}\underset{n-1}{\downarrow}\underset{n}{\downarrow}\underset{n+1}{\downarrow}\underset{n+2}{\downarrow}\uparrow\uparrow\uparrow\ldots$$

$\to$ Flip all spins, from $-\infty$ to site <u>n</u>, down

Energy cost $=$ <u>1</u> bond $(n,n+1)$ only $\Rightarrow \Delta E_{dw} = 2J.$

$$\underset{n}{\downarrow}\underset{n+1}{\uparrow}$$

Denote state with 1 domain wall by the mid-point of the bond $(n, n+1)$ separating $\downarrow$ spins from $\uparrow$ spins $\Rightarrow$ $|\tilde{n}\rangle$, where $\tilde{n} = n + \frac{1}{2}$



(Sites $\tilde{n}$ form dual lattice)

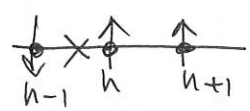
For $g=0$ domain wall does not move.

Now the limit $g \to 0$, but finite:

$$H|\tilde{n}\rangle = -J\sum_i \sigma_i^z \sigma_{i+1}^z |\tilde{n}\rangle - Jg\sum_i \sigma_i^x |\tilde{n}\rangle$$

$$\hat{\sigma}_n^x \left|\ldots \underset{\downarrow}{\varphi_{2n-1}} \; \underset{\downarrow}{\varphi_{2n}} \; \underset{\uparrow}{\varphi_{1n+1}} \; \underset{\uparrow}{\varphi_{1n+2}} \ldots\right\rangle = \left|\varphi_{2,n-1} \; \varphi_{1,n} \; \varphi_{1,n+1} \ldots\right\rangle$$

$$= \quad = |\widetilde{n-1}\rangle$$



$\Rightarrow$ Domain wall moved to the left.

Clearly, $\hat{\sigma}_{n+1}^x |\downarrow \underset{n}{\downarrow} \underset{n+1}{\uparrow} \uparrow\rangle = |\downarrow \underset{n}{\downarrow} \underset{n+1}{\downarrow} \underset{n+2}{\uparrow}\rangle = |\widetilde{n+1}\rangle$

moves DW to the right.

(Note that $\hat{\sigma}_j^x$ acting away from the domain wall also flips spin $\Rightarrow$ and this increases the energy by $4J$ $\rightarrow$ such a process increases the energy.)

Hence $(\hat{H} - E_0)|\tilde{n}\rangle = 2J|\tilde{n}\rangle - gJ\left(|\widetilde{n-1}\rangle + |\widetilde{n+1}\rangle\right)$.

$$\Rightarrow \varepsilon(k) = 2J - 2gJ\cos k = 2J(1 - g\cos k).$$

Ordered phase ($g \ll 1$)

Operator creating domain wall on $\tilde{n} = n + \frac{1}{2}$

$$\mu^z_{\tilde{n}} = \prod_{j \leq n} \sigma^x_j$$

(assuming that the GS is $|0_\uparrow\rangle = |\uparrow\uparrow\uparrow\ldots\rangle$ which can be achieved by fixing the first spin to be $\uparrow$)

(Non-local operator)

$$\mu^z_{\tilde{n}} |0_\uparrow\rangle = |\downarrow\downarrow\downarrow\underset{n}{\downarrow}\overset{\tilde{n}}{\times}\underset{n+1}{\uparrow}\uparrow\uparrow\rangle = |\tilde{n}\rangle$$

Clearly $\langle 0_\uparrow | \mu^z_{\tilde{n}} | 0_\uparrow \rangle = \langle 0_\uparrow | \tilde{n} \rangle = 0.$

Disordered phase ($g \gg 1$)

GS $|0\rangle = |\rightarrow\rightarrow\rightarrow\rightarrow\rangle$

Remember, $\hat{\sigma}^x |\rightarrow\rangle = |\rightarrow\rangle$

Hence, $\langle 0 | \mu^z_{\tilde{n}} | 0 \rangle = \langle \rightarrow\rightarrow\rightarrow\ldots | \prod_{j \leq n} \sigma^x_j | \rightarrow\rightarrow\rightarrow\ldots \rangle$

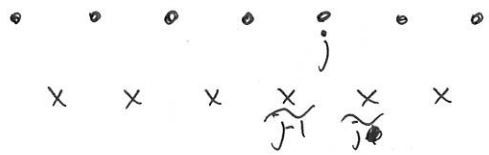
$$= 1$$

$\Rightarrow$ Disordered phase is a condensate of domain walls which proliferated beyond $g = 1$ point!

What is $\mu^x_{\tilde{n}}$? By analogy with $g \gg 1$ situation, it should be an operator that <u>moves domain wall</u>

$$\Rightarrow \mu^x_{\tilde{n}} = \sigma^z_n \sigma^z_{n+1} \; !$$

Duality



original lattice

dual lattice

$$(\mu^x_{\tilde{n}})^2 = 1 = (\mu^z_{\tilde{n}})^2$$

$$(\sigma^x_n)^2 = 1 = (\sigma^z_n)^2$$

Observe that $\sigma^x_j = \mu^z_{\widetilde{j-1}}\, \mu^z_{\tilde{j}} = \left(\prod_{i \le j-1} \sigma^x_i\right) \cdot$

$$\cdot \left(\prod_{k \le j} \sigma^x_k\right) = \left(\prod_{i \le j-1} (\sigma^x_i)^2\right) \sigma^x_j = \sigma^x_j .$$

$$\Rightarrow H = -J\left(\sum_i \sigma^z_i \sigma^z_{i+1} + g \sum_i \sigma^x_i\right) =$$

$$= -J\left(\sum_{\tilde{i}} \mu^x_{\tilde{i}} + g \sum_{\tilde{i}} \mu^z_{\widetilde{i-1}}\, \mu^z_{\tilde{i}}\right) =$$

$$= -Jg\left(\sum_{\tilde{i}} \mu^z_{\widetilde{i-1}}\, \mu^z_{\tilde{i}} + \frac{1}{g} \sum_{\tilde{i}} \mu^x_{\tilde{i}}\right)$$

$\Rightarrow$ $g=1$ is a <u>self-dual</u> point

This proves that quantum phase transition takes place at this point.

$g=0$ ordered | $\boxed{g=1}$ | disordered $g=\infty$

$\uparrow\uparrow\uparrow\uparrow$

$\downarrow\downarrow\downarrow\downarrow$

$\downarrow\downarrow\downarrow\uparrow\uparrow\uparrow$

domain wall

$\varepsilon(k) = 2J \cdot (1 - g\cos k)$

$\varepsilon_{min} = 2J(1-g)$

$\rightarrow \rightarrow \rightarrow \rightarrow \rightarrow$

$\rightarrow \rightarrow \leftarrow \rightarrow \rightarrow$

spin flip

$\varepsilon(k) = 2J(g - \cos k)$

$\varepsilon_{min} = 2J(g-1)$

Note that at $g=1$ the minimal energy of creating the excitation $\varepsilon_{min} = 0 \Rightarrow$ this is the point of the <u>quantum phase transition</u>.

$\{\sigma^x, \sigma^y, \sigma^z\}$ Pauli matrices

$\sigma^x \sigma^y = i\sigma^z \ldots \Rightarrow \sigma^i \sigma^j = i\,\varepsilon^{ijk}\,\sigma^k$

$(i,j,k) \Leftrightarrow (x,y,z)$

$\Rightarrow \sigma^y \sigma^x = -i\sigma^z$

$\Rightarrow \quad \sigma^i \sigma^j + \sigma^j \sigma^i = 2\,\delta_{ij}$

$(\sigma^i)^2 = 1 \quad \forall i$

Emergence of fermions

$\cancel{\sigma^x \quad \mu^z \quad n \quad \tilde{n}}$

<u>Define</u> $\chi_i = \frac{1}{2}(c + c^+) = \sigma_i^x \underbrace{\prod_{j<i} \sigma_j^z}$

$\bar{\chi}_i = \sigma_i^y \prod_{j<i} \sigma_j^z$

$\sigma_1^z \sigma_2^z \sigma_3^z \sigma_4^z \ldots =$

$$\chi_i \chi_k = \sigma_i^x \prod^{i-1} \sigma_j^z \; \sigma_k^x \prod^{k-1} \sigma^z$$

$$= \sigma_i^x \sigma_{i-1}^z \sigma_{i-2}^z \ldots \underbrace{\sigma_k^z \sigma_k^x}_{i\sigma_k^y}$$

while $\chi_k \chi_i = \sigma_k^x \prod^{k-1} \sigma^z \; \sigma_i^x \prod^{i-1} \sigma_j^z =$

$$= \sigma_k^x \sigma_i^x \sigma_{i-1}^z \sigma_{i-2}^z \ldots \sigma_k^z =$$

$$= \sigma_i^x \sigma_{i-1}^z \sigma_{i-2}^z \underbrace{\sigma_k^x \sigma_k^z}_{-i\sigma_k^y}$$

$\Rightarrow \chi_i \chi_k = -\chi_k \chi_i$ for $k \neq i$

χ's <u>anti-commute</u>!

But $(\chi_i)^2 = (\sigma_i^x \sigma_{i-1}^z \sigma_{i-2}^z \ldots)^2 = 1.$

Also, $\bar{\chi}_i \bar{\chi}_k = \sigma_i^y \sigma_{i-1}^z \sigma_{i-2}^z \ldots \underbrace{\sigma_k^z \sigma_k^y}_{i\sigma_k^x = -i\sigma_k^x}$

while $\bar{\chi}_k \bar{\chi}_i = \sigma_i^y \sigma_{i-1}^z \sigma_{i-2}^z \underbrace{\sigma_k^y \sigma_k^z}_{i\sigma_k^x}$

$\bar{\chi}_i \bar{\chi}_k = -\bar{\chi}_k \bar{\chi}_i$

$$\chi_i \bar{\chi}_k = \sigma_i^x \prod^{i-1} \sigma_j^z \; \sigma_k^y \prod^{k-1} \sigma_n^z = \sigma_i^x \, \sigma_{i-1}^z \, \sigma_{i-2}^z \cdots \underbrace{\sigma_k^z \, \sigma_k^y}_{i\varepsilon^{zyx}\sigma_k^x}$$

$$\bar{\chi}_k \chi_i = \sigma_k^y \prod^{k-1} \sigma_n^z \; \sigma_i^x \, \sigma_{i-1}^z \, \sigma_{i-2}^z \cdots \sigma_k^z \prod^{k-1} \sigma_j^z =$$

$$= \sigma_{i-1}^z \, \sigma_{i-2}^z \cdots \underbrace{\sigma_k^y \, \sigma_k^z}_{i\varepsilon^{yzx}\sigma_k^x = i\sigma_k^x} \sigma_i^x = -\chi_i \bar{\chi}_k$$

Thus $\{\chi_i, \chi_k\} = 2\delta_{ik}$, $\{\chi_i, \bar{\chi}_k\} = 0$

let $c_i = \frac{1}{2}(\chi_i + i\bar{\chi}_i)$

Then $\{c_i, c_k\} = \frac{1}{4}\{\chi_i + i\bar{\chi}_i, \chi_k + i\bar{\chi}_k\} =$

$$= \frac{1}{4}\{\chi_i, \chi_k\} + i\{\chi_i, \bar{\chi}_k\} + i\{\bar{\chi}_i, \chi_k\} - \{\bar{\chi}_i, \bar{\chi}_k\} = 0$$

But $\{c_i, c_k^+\} = \frac{1}{4}\{\chi_i + i\bar{\chi}_i, \chi_k - i\bar{\chi}_k\} =$

$$= \frac{1}{4}\Big(\underbrace{\{\chi_i, \chi_k\}}_{2\delta_{ik}} + \underbrace{\{\bar{\chi}_i, \bar{\chi}_k\}}_{2\delta_{ik}} + i\underbrace{\{\bar{\chi}_i, \chi_k\}}_{2\delta_{ik}} - i\underbrace{\{\chi_i, \bar{\chi}_k\}}_{2\delta_{ik}}\Big)$$

$$= \delta_{ik}$$

Thus $c_i = \frac{1}{2}(\chi_i + i\bar{\chi}_i) = \frac{1}{2}(\sigma_i^x + i\sigma_i^y) \prod^{i-1} \sigma_j^z =$

$$= \frac{1}{2}\hat{\sigma}_i^+ \prod_{j<i} \sigma_j^z$$

$$c_i^+ = \frac{1}{2}\hat{\sigma}_i^- \prod_{j<i} \sigma_j^z$$

are fermion operators!

It is actually more convenient to work with the <u>dual</u> Hamiltonian

$$H = -J \sum_{\tilde{i}} \left(g\, \mu^z_{\tilde{i}-1} \mu^z_{\tilde{i}} + \mu^x_{\tilde{i}} \right)$$

which is defined on medial lattice $\tilde{i} = i + \frac{1}{2}$.

This is the lattice on which domain walls naturally live $\Rightarrow$ below we denote its sites by usual (i) but will express Majorana fermions χ_i and $\bar{\chi}_i$ in terms of "disorder" variables μ^x_i and μ^z_i.

Properties of $\{\chi\}$:

$\underline{\underline{l<i}}$ $\quad \chi_i \chi_l = \mu_i^x (\mu_{i-1}^z \mu_{i-2}^z \cdots \mu_l^z \mu_{l-1}^z \cdots) \mu_l^x (\mu_{l-1}^z \cdots) =$

$$= \mu_i^x (\mu_{i-1}^z \cdots \mu_{l+1}^z) \underline{\underline{\mu_l^z \mu_l^x}} \underbrace{(\mu_{l-1}^z \cdots)^2}_{1}$$

$$\chi_l \chi_i = \mu_l^x (\mu_{l-1}^z \cdots) \mu_i^x (\mu_{i-1}^z \cdots \mu_{l+1}^z \mu_l^z \mu_{l-1}^z \cdots) =$$

$$= \mu_l^x \mu_i^x \mu_l^z (\mu_{l-1}^z \cdots \mu_{l+1}^z) =$$

$$= \underline{\underline{\mu_l^x \mu_l^z}} \mu_i^x (\mu_{i-1}^z \cdots \mu_{l+1}^z) = -\chi_i \chi_l \,!$$

Since $\mu_l^x \mu_l^z = -i \mu_l^y = -(\mu_l^z \mu_l^x) = -(i \mu_l^y)$
we find that $\chi_i \chi_l = -\chi_l \chi_i$

While at the same site $(\chi_i)^2 = 1$.

Now, $\chi_i \bar{\chi}_l = \mu_i^x (\mu_{i-1}^z \cdots \mu_{l+1}^z) \mu_l^z \mu_l^y$

$$\bar{\chi}_l \chi_i = \mu_l^y \mu_i^x (\mu_{i-1}^z \cdots \mu_{l+1}^z \mu_l^z) =$$

$$= \mu_l^y \mu_i^x \mu_l^z (\mu_{i-1}^z \cdots \mu_{l+1}^z)$$

$\underline{\text{For } l<i}$: $\quad \bar{\chi}_l \chi_i = \mu_i^x \underbrace{\mu_l^y \mu_l^z}_{i\mu_l^x} (\mu_{i-1}^z \cdots \mu_{l+1}^z)$

$$\chi_i \bar{\chi}_l = \mu_i^x \underbrace{\mu_l^z \mu_l^y}_{-i\mu_l^x} (\mu_{i-1}^z \cdots \mu_{l+1}^z) = -\bar{\chi}_l \chi_i$$

$\underline{\text{Same site}}$ $\underline{\underline{l=i}}$: $\quad \chi_i \bar{\chi}_i = \mu_i^x \mu_i^y = i \mu_i^z$

$$\bar{\chi}_i \chi_i = \mu_i^y \mu_i^x = -i \mu_i^z$$

Thus, $\{\chi_i, \chi_l\} = \chi_i \chi_l + \chi_l \chi_i = 2\delta_{il}$

$$\{\bar{\chi}_i, \bar{\chi}_l\} = 2\delta_{il}$$

while $\{\chi_i, \bar{\chi}_l\} = 0$ always.

Like Majorana fermions anticommute, except for $i=l$, while unlike Majorana fermions always anti-commute.

For usual fermions: $\begin{cases} c_i = \frac{1}{2}(\chi_i + i\bar{\chi}_i) \\ c_i^+ = \frac{1}{2}(\chi_i - i\bar{\chi}_i) \end{cases}$

$$\{c_i, c_j^+\} = \frac{1}{4}\{\chi_i + i\bar{\chi}_i, \chi_j - i\bar{\chi}_j\} =$$

$$= \frac{1}{4}\Big(\underbrace{\{\chi_i, \chi_j\}}_{2\delta_{ij}} + \underbrace{\{\bar{\chi}_i, \bar{\chi}_j\}}_{2\delta_{ij}} - i\underbrace{\{\chi_i, \bar{\chi}_j\}}_{0} + i\underbrace{\{\bar{\chi}_i, \chi_j\}}_{0}\Big) = \delta_{ij}$$

$$\{c_i, c_j\} = \frac{1}{4}\Big(\{\chi_i, \chi_j\} - \{\bar{\chi}_i, \bar{\chi}_j\}\Big) = 0.$$

Inverse: $\begin{cases} \chi_i = c_i + c_i^+ \\ \bar{\chi}_i = -i(c_i - c_i^+) \end{cases}$

$$\chi_i = \mu_i^x \prod_{j<i} \mu_j^z \,, \quad \bar{\chi}_i = \mu_i^y \prod_{j<i} \mu_j^z$$

$$\chi_i^2 = \bar{\chi}_i^2 = 1$$

$$\bar{\chi}_i \chi_i = \mu_i^y \prod_{j<i} \mu_j^z \; \mu_i^x \prod \mu_j^z = -i \mu_i^z$$

$$\Rightarrow \mu_i^z = i \bar{\chi}_i \chi_i$$

$$\bar{\chi}_i \chi_{i+1} = \mu_i^y \left(\prod_{j<i} \mu_j^z\right) \mu_{i+1}^x \left(\prod_{j<i+1} \mu_j^z\right) =$$

$$= \mu_i^y \left(\mu_{i-1}^z \mu_{i-2}^z \cdots\right) \mu_{i+1}^x \left(\mu_i^z \mu_{i-1}^z \mu_{i-2}^z \cdots\right) =$$

$$= \underbrace{\mu_i^y \mu_i^z}_{i \varepsilon^{yzx} \mu_i^x} \mu_{i+1}^x \left(\mu_{i-1}^z \mu_{i-2}^z \cdots\right)^2 = i \mu_i^x \mu_{i+1}^x$$

$$\Rightarrow \mu_i^x \mu_{i+1}^x = -i \bar{\chi}_i \chi_{i+1}$$

Thus, $H = -J \sum_i \left(-ig \bar{\chi}_i \chi_{i+1} + i \bar{\chi}_i \chi_i\right) =$

$$= iJ \sum_i \left(g \bar{\chi}_i \chi_{i+1} - \bar{\chi}_i \chi_i\right)$$

Express H in c, c^+:

$$H = J \sum_i \left(g (c_i - c_i^+)(c_{i+1} + c_{i+1}^+) - (c_i - c_i^+)(c_i + c_i^+) \right)$$

$$= J \sum_i \left(g\, c_i c_{i+1} + c_i c_{i+1}^+ - c_i^+ c_{i+1} - c_i^+ c_{i+1}^+ \right.$$
$$\left. - c_i c_i^+ + c_i^+ c_i \right)$$

But $c_i c_i^+ = 1 - c_i^+ c_i$ (and we can drop 1)

$$H = J \sum_i \left(2 c_i^+ c_i - g \left(c_i^+ c_{i+1} + c_{i+1}^+ c_i + c_{i+1} c_i + c_i^+ c_{i+1}^+ \right) \right)$$

Fourier: $c_i = \frac{1}{\sqrt{N}} \sum_k e^{ikr_i} c_k$

$$c_i^+ = \frac{1}{\sqrt{N}} \sum_k e^{-ikr_i} c_k^+$$

$$H = \frac{J}{N} \sum_i \sum_{k_1 k_2} \Big\{ 2 c_{k_1}^+ c_{k_2} e^{-ik_1 r_i + ik_2 r_i}$$

$$- g \Big[\Big(e^{-ik_1 r_i + ik_2 (r_i + 1)} c_{k_1}^+ c_{k_2} + c_{k_1} c_{k_2} e^{ik_1 (r_i+1) + ik_2 r_i}$$
$$+ e^{-ik_1 (r_i+1) + ik_2 r_i} \Big) \qquad k_1 = -k_2$$

$$+ c_{k_1}^+ c_{k_2}^+ e^{-ik_1 r_i - ik_2 (r_i + 1)} \Big] \Big\} =$$

$$= J \sum_k \Big\{ 2 c_k^+ c_k - g \Big[(e^{ik} + e^{-ik}) c_k^+ c_k + c_k c_{-k} e^{ik}$$
$$+ c_k^+ c_{-k}^+ e^{ik} \Big] \Big\}$$

$$= J \sum_k \Big\{ 2 (1 - g \cos k) c_k^+ c_k - i g \sin k \left(c_k c_{-k} + c_k^+ c_{-k}^+ \right) \Big\}$$

because $\sum_k \cos k \, c_k^+ c_{-k}^+ = 0$.

H written in terms of fermions c_k^+, c_k does not conserve their numbers: terms $c_k^+ c_{-k}^+$ and $c_k c_{-k}$ describe creation and annihilation of the particles in pairs.

Bogoliubov transformation

We seek
$$\eta_k = u_k c_k - i v_k c_{-k}^+$$
$$\eta_k^+ = u_k c_k^+ + i v_k c_{-k}$$

with $u_k = u_{-k}$, $v_k = -v_{-k}$

Require η's to be fermions $\Rightarrow \{\eta_k, \eta_p^+\} = \delta_{kp}$

$$\Rightarrow \{u_k c_k - i v_k c_{-k}^+, u_p c_p^+ + i v_p c_{-p}\} =$$

$$= u_k u_p \underbrace{\{c_k, c_p^+\}}_{\delta_{kp}} + v_k v_p \underbrace{\{c_{-k}^+, c_{-p}\}}_{\delta_{k,-p}} + i u_k v_p \underbrace{\{c_k, c_{-p}\}}_{=0}$$

$$- i v_k u_p \underbrace{\{c_{-k}^+, c_p^+\}}_{=0} = (u_k u_p + v_k v_p)\,\delta_{kp} =$$

$$= (u_k^2 + v_k^2)\,\delta_{kp} \Rightarrow \boxed{u_k^2 + v_k^2 = 1}$$

Also check $\{\eta_k, \eta_p\} = \{u_k c_k - i v_k c_{-k}^+, u_p c_p - i v_p c_{-p}^+\} =$

$$= -i u_k v_p \underbrace{\{c_k, c_{-p}^+\}}_{\delta_{k,-p}} - i v_k u_p \underbrace{\{c_{-k}^+, c_p\}}_{\delta_{-k,p}} =$$

$$= (-i)\,(u_k \underbrace{v_{-k}}_{=-v_k} + u_{-k} v_k) = 0$$

Inverse: $\begin{cases} \eta^+_{-k} = u_{-k} c^+_{-k} + i v_{-k} c_k = u_k c^+_{-k} - i v_k c_k \\ \eta_k = u_k c_k - i v_k c^+_{-k} \end{cases}$

Thus $u_k \eta_k + i v_k \eta^+_{-k} = u_k^2 c_k + v_k^2 c_k = c_k$.

$$\Rightarrow \begin{cases} c^+_k = u_k \eta^+_k - i v_k \eta_{-k} \\ c^+_{-k} = u_k \eta^+_{-k} + i v_k \eta_k \end{cases}$$

Up to a constant

$$H = J \sum_k (c^+_k, c_{-k}) \begin{pmatrix} 1 - g\cos k & -ig\sin k \\ ig\sin k & -(1-g\cos k) \end{pmatrix} \begin{pmatrix} c_k \\ c^+_{-k} \end{pmatrix} =$$

$$= J \sum_k (c^+_k, c_{-k}) \begin{pmatrix} (1-g\cos k) c_k - ig\sin k\, c^+_{-k} \\ ig\sin k\, c_k - (1-g\cos k) c^+_{-k} \end{pmatrix} =$$

$$= J \sum_k (1-g\cos k)(c^+_k c_k - \underbrace{c_{-k} c^+_{-k}}_{1 - c^+_{-k} c_{-k}}) - ig\sin k\, c^+_k c^+_{-k}$$

$$+ ig\sin k \underbrace{c_{-k} c_k}_{-c_k c_{-k}}$$

$$\begin{pmatrix} c_k \\ c^+_{-k} \end{pmatrix} = \begin{pmatrix} u_k & i v_k \\ i v_k & u_k \end{pmatrix} \begin{pmatrix} \eta_k \\ \eta^+_{-k} \end{pmatrix}$$

h.c. = transpose and conjugate

$$(c^+_k, c_{-k}) = \begin{pmatrix} c_k \\ c^+_{-k} \end{pmatrix}^+ = (\eta^+_k, \eta_{-k}) \begin{pmatrix} u_k & i v_k \\ i v_k & u_k \end{pmatrix}^+ =$$

$$= (\eta^+_k, \eta_{-k}) \begin{pmatrix} u_k & -i v_k \\ -i v_k & u_k \end{pmatrix} = (u_k \eta^+_k - i v_k \eta_{-k},$$

$$-i v_k \eta^+_k + u_k \eta_{-k})$$

Hence $H = J \sum_k (\eta_k^+, \eta_{-k}) \begin{pmatrix} u_k & -iv_k \\ -iv_k & u_k \end{pmatrix} \begin{pmatrix} 1-g\cos K & -ig\sin K \\ ig\sin K & -(1-g\cos K) \end{pmatrix} \begin{pmatrix} u_k & iv_k \\ iv_k & u_k \end{pmatrix} \begin{pmatrix} \eta_k \\ \eta_{-k}^+ \end{pmatrix}$

$$= J \sum_k (\eta_k^+, \eta_{-k}) \begin{bmatrix} (u_k^2 - v_k^2)(1-g\cos K) + 2u_k v_k g\sin K; & i\left[2u_k v_k(1-g\cos K) - g\sin K(u_k^2 - v_k^2)\right] \\ i\left[-2u_k v_k(1-g\cos K) + g\sin K(u_k^2 - v_k^2)\right]; & (-)(u_k^2 - v_k^2)(1-g\cos K) - 2u_k v_k g\sin K \end{bmatrix} \begin{pmatrix} \eta_k \\ \eta_{-k}^+ \end{pmatrix}$$

We require of-diagonal elements to be <u>zero</u>

$$\Rightarrow \quad 2u_k v_k (1-g\cos K) = (u_k^2 - v_k^2)\, g\sin K$$

let $u_k = \cos\theta_k$, $v_k = \sin\theta_k$ (NOTE: $u_k^2 + v_k^2 = 1$ as it should be)

$$\Rightarrow \quad \sin(2\theta_k)(1-g\cos K) = \cos(2\theta_k)\, g\sin K$$

$$\tan(2\theta_k) = \frac{g\sin K}{1-g\cos K}$$

Diagonal element (11):

$$\cos(2\theta_k)(1-g\cos K) + \sin(2\theta_k)\, g\sin K =$$

$$= \cos(2\theta_k)\Big[1 - g\cos K + g\sin K \underbrace{\left(\frac{g\sin K}{1-g\cos K}\right)}_{\tan(2\theta_k)}\Big] =$$

$$= \cos(2\theta_k)\, \frac{(1-g\cos K)^2 + (g\sin K)^2}{1-g\cos K} = \sqrt{(1-g\cos K)^2 + g^2\sin^2 K}$$

But $\cos(2\theta_k) = \dfrac{1}{\sqrt{1+\tan^2 2\theta_k}} = \dfrac{1-g\cos K}{\sqrt{(1-g\cos K)^2 + g^2\sin^2 K}}$

Thus $H = J \sum_k (\eta_k^+, \eta_{-k}) \begin{pmatrix} \omega_k & 0 \\ 0 & -\omega_k \end{pmatrix} \begin{pmatrix} \eta_k \\ \eta_{-k}^+ \end{pmatrix} =$

$$= J \sum_k \omega_k \eta_k^+ \eta_k - \omega_k \underbrace{\eta_{-k} \eta_{-k}^+}_{1 - \eta_{-k}^+ \eta_{-k}} =$$

$$= J \sum_k \omega_k \left(\eta_k^+ \eta_k + \eta_{-k}^+ \eta_{-k} \right) - \omega_k$$

Since $\omega_k = \omega_{-k}$

$$H = 2J \sum_k \omega_k \eta_k^+ \eta_{-k} .$$

Where $\omega_k = \sqrt{(1 - g\cos k)^2 + g^2 \sin^2 k} = \sqrt{1 + g^2 - 2g\cos k}$

The spectrum is gapped everywhere <u>except for $g=1$</u>

$$\omega_k (g=1) = \sqrt{2 - 2\cos k} = \sqrt{4 \sin^2 \tfrac{k}{2}} = 2 \left| \sin \tfrac{k}{2} \right| .$$

At $g=1$ the spectrum is gapless at <u>$k=0 \Rightarrow$ critical point!</u>

Notice: excitations are fermions.